LEDGER · 2026-09-02 · SEMICONDUCTOR PACKAGING / POWER ELECTRONICS

Pressureless Micro-Silver-Indium (μAg-In) TLP Sintering Paste via Mechanical Alloying

CLEARED

This concept cleared all seven gates and every voiding sub-test.

Incumbent benchmark	MacDermid Alpha Argomax 2148
Entry application	Die-attachment of SiC MOSFETs to Direct Bonded Copper substrates in EV traction inverters
Measured advantage	>2.6x die shear strength after 1000h at 300°C under pressureless conditions (53 MPa vs <20 MPa for incumbent MacDermid Argomax 2148)

Time to first revenue	9 months
Gross margin at parity	73.4%
Startup CapEx	$150,000
Payback from first sale	0.1 months
Capital productivity	360.0x
Regulatory risk	Moderate — qualification costs reported ($120,000 estimated)

Assessed 2026-09-02 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts